

INTRODUCTION TO MACHINE LEARNING

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that enables computers to learn patterns from data and make decisions without being explicitly programmed.

TYPES OF MACHINE LEARNING

1. Supervised Learning Supervised learning uses labeled datasets to train models. Examples include email spam detection and house price prediction.
2. Unsupervised Learning Unsupervised learning works with unlabeled data. It is commonly used for clustering and pattern discovery.
3. Reinforcement Learning Reinforcement learning involves an agent learning through rewards and penalties. It is widely used in robotics and game-playing AI.

APPLICATIONS OF MACHINE LEARNING

Machine Learning is used in:

- Healthcare for disease prediction
- Finance for fraud detection
- E-commerce for recommendation systems
- Transportation for self-driving vehicles

ADVANTAGES

- Automates decision-making
- Handles large datasets efficiently
- Improves prediction accuracy

CHALLENGES

- Requires large amounts of quality data
- Can be computationally expensive
- Models may suffer from bias

CONCLUSION

Machine Learning has become one of the most important technologies in modern computing. Its applications continue to grow across industries, making it a crucial field for future innovation.